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We have two goals for these two talks:

(1) Understand the statement of the geometric expansion of the local trace formula.

(2) See how the geometric expansion is equivalent to the multiplicity formula.

1. Recollections

Definition 1.1. For $x \in G_{ss}(F)$, We consider the determinant $D^G(x) := |\det(1 - \mathrm{Ad}(x))|_{\mathfrak{g}/\mathfrak{g}_x}|$.

Recall from Xin's: A quasi-character $\theta \in QC(G)$ in the p-adic case is (by definition) locally[1]

$$\theta(xe^Y) = \sum_{\mathcal{O} \in \mathrm{Nil}(\mathfrak{g}_x)} c_{\theta,\mathcal{O}}(x)\hat{j}_{\mathcal{O}}(Y), \quad \text{for } Y \in \omega_{x,reg}$$

where $\hat{j}_{\mathcal{O}} \in C^\infty_{loc.int.}(\mathfrak{g}_{reg})$ are the Fourier transforms of orbital integrals.

Definition 1.2. For $\theta \in QC(G)$, we define $c_\theta(x) = \frac{1}{\#\mathrm{Nil}_{reg}(\mathfrak{g}_x)} \sum_{\mathcal{O} \in \mathrm{Nil}_{reg}(\mathfrak{g}_x)} c_{\theta,\mathcal{O}}(x)$.

Proposition 1.3. *We have $c_\theta(x) = 0$ unless G_x is quasi-split. If G_x is quasi split, then*

$$D^G(x)^{1/2} c_\theta(x) = \frac{1}{\#W(G_x, T_{qs,x})} \lim_{\substack{x' \in T_{qs,x} \\ x' \to x}} D^G(x')^{1/2}\theta(x').$$

2. The picture of m_{geom}

Definition 2.1. For $x \in H_{ss}(F)$, $\Delta(x) := D^G(x)D^H(x)^{-2}$.

We will define a linear functional $m_{geom,s} \colon QC(G(F)) \to \mathbb{C}$ to have the shape

$$m_{geom,s}(\theta) := \int_{\Gamma(G,H)} D^G(x)^{1/2} c_\theta(x) \Delta(x)^{s-1/2} \mathrm{d}x$$

and prove it is absolutely convergent for $\mathrm{Re}(s) > 0$. Here, $\Gamma(G,H)$ is a certain subset of semisimple conjugacy clases of $H(F)$. We will then prove that the limit $m_{geom}(\theta) := \lim_{s \to 0^+} m_{geom,s}(\theta)$ exist:

[1] In the real case, we only have an assymptotic expansion $\theta(xe^Y) = \sum_{\mathcal{O} \in \mathrm{Nil}_{reg}(\mathfrak{g}_x)} c_{\theta,\mathcal{O}}(x)\hat{j}_{\mathcal{O}}(Y) + O(|Y|)$. This is also true in the p-adic case since $\hat{j}_{\mathcal{O}}(Y) = O(|Y|)$ if $\mathcal{O}$ is not regular.

Theorem 2.2. *For each $\theta \in QC(G(F))$ the limit $m_{geom}(\theta) := \lim_{s\to 0^+} m_{geom,s}(\theta)$ exists, and m_{geom} defines a continuous functional on $QC(G(F))$.*

Remark 2.3. We should think of this as a "regularized integral"

$$m_{geom}(\theta) = \int_{\Gamma(G,H)}^{*} D^H(x) c_\theta(x) \mathrm{d}x.$$

Eventually we will have the following theorems

Theorem 2.4. *$J(f) = J_{geom}(f) := m_{geom}(\theta_f)$ for all strongly cuspidal $f \in \mathcal{C}_{scusp}(G(F))$.*

Theorem 2.5. *$m(\pi) = m_{geom}(\pi) := m_{geom}(\theta_\pi)$ for all tempered π.*

3. Spaces of conjugacy classes

Let's talk a bit about GGP triples (G, H, ξ). These come from a pair (V, W) where $V = W \oplus Z$ with Z odd dimensional and quasi-split. Then $G = U(V) \times U(W)$ and we have $H = U(W) \ltimes N$ where N is as follows: we have a parabolic P_V of $U(V)$ given by

$$\begin{pmatrix} U(W) & * & * & \cdots & * & * \\ 0 & * & * & \cdots & * & * \\ 0 & 0 & * & \cdots & * & * \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & * & * \\ 0 & 0 & 0 & \cdots & 0 & * \end{pmatrix}$$

and N is its unipotent. Then $U(W)$ acts on N, and the character ξ comes from $\lambda\colon N \to \mathbb{G}_a$ given by summing over the one above diagonal in the Z part, which extends to $H \twoheadrightarrow N \to \mathbb{G}_a$.

We will now define $\Gamma(G, H)$ as subset of semisimple conjugacy classes of H. For $x \in H_{ss}(F)$, we have a triple $(G_x(F), H_x(F), \xi_x = \xi|_{H_x(F)})$. It is not quite a GGP triple, but it is close to one, as we will explain. By $H(F)$-conjugation, we may assume $x \in U(W)$. We consider $W = W'_x \oplus W''_x$ and $V = V'_x \oplus V''_x$ where V'_x, W'_x are the kernels of $1 - x$, and the complements are the image of $1 - x$. Note that

$$U(V)_x = U(V'_x) \times U(V''_x)_x, \quad U(W)_x = U(W'_x) \times U(W''_x)_x,$$

and so we have a natural decomposition

$$G_x = G'_x \times G''_x, \quad G'_x := U(V'_x) \times U(W'_x), \quad G''_x := U(V''_x)_x \times U(W''_x)_x.$$

as well as

$$H_x = H'_x \times H''_x, \quad H'_x := U(W'_x) \ltimes N_x, \quad H''_x := U(W''_x)_x.$$

This is because $U(W''_x)_x$ commutes with N_x: N_x is concentrated in the first row in the matrix picture, so $N_x = \{n \colon xn = n\}$ (doing matrix multiplication under that basis); but now $U(W''_x)$ is precisely h's such that $xv = v \implies hv = v$.

So in fact

$$(G_x, H_x, \xi_x) = (G'_x, H'_x, \xi'_x) \times (G''_x, H''_x, 1)$$

where (G'_x, H'_x, ξ'_x) is the GGP triple of W'_x, V'_x. Note $Z \subseteq V'_x$, and so $V''_x = W''_x$, and $H''_x \hookrightarrow G''_x$ is the diagonal embedding. Raphael calls $(G''_x, H''_x, 1)$ an *Arthur triple*.

Remark 3.1. Given an element of $\Gamma(H)$, the above decomposition is well-defined up inner automorphisms.

Definition 3.2 ([BP20, Page 259])**.** We define $\Gamma(G, H) \subseteq \Gamma(H)$ to be the subset of x such that H''_x (and hence G''_x) are anisotropic tori.

To give a more concrete description of $\Gamma(G, H)$ and to equip it with a measure, we first parametrize break it down in terms of the anisotropic tori H''_x.

Definition 3.3 ([BP20, Page 259])**.** We denote $\underline{\mathcal{T}}$ the set of subtori of $U(W)$ where $T \in \underline{\mathcal{T}}$ if there exists a nondegenerate $W'' \subseteq W$ such that T is a maximal anisotropic subtorus of $U(W'') \subseteq U(W)$.

Now given T, we characterize the possible x:

Definition 3.4 ([BP20, Page 259])**.** For $T \in \underline{\mathcal{T}}$, we consider $T_\natural(F)$ the Zariski open subset of regular elements $x \in T(F)$ without eigenvalue 1 on W''.

Proposition 3.5 ([BP20, Equation (11.1.3)])**.** *$\Gamma(G, H)$ is the set of conjugacy classes that meet $\bigcup_{T \in \underline{\mathcal{T}}} T_\natural(F)$.*

Proof. By definitions, any conjugacy class of $\Gamma(G, H)$ meets it. Conversely, if $T \in \underline{\mathcal{T}}$ and $x \in T_\natural(F)$, then W'' has an eigenspace decomposition for x, and so $W'' \subseteq W''_x$. But $x \in U(W'')$, so $W'' = W''_x$, and thus $T = U(W'')_x = U(W''_x)_x = H''_x$. □

Note that by the above W'' is uniquely determined by T, and we denote $W(T) := W(U(W''_T), T)$. Fixing $\mathcal{T} \subseteq \underline{\mathcal{T}}$ a (finite!) set of representatives under the $U(W)(F)$-action, we get

$$\Gamma(G,H) \simeq \bigsqcup_{T\in\mathcal{T}} T_\natural(F)/W(T).$$

We use this to equip $\Gamma(G,H)$ with a topology and a measure: for $\varphi \in C(\Gamma(G,H))$, we define

$$\int_{\Gamma(G,H)} \varphi(x)\mathrm{d}x := \sum_{T\in\mathcal{T}} \frac{1}{\#W(T)} \int_{T(F)} \varphi(t)\mathrm{d}t$$

where $\int_{T(F)}$ is normalized to have total mass 1.

4. Convergence of $m_{geom,s}$ when $\mathrm{Re}(s) > 0$

Proposition 4.1 ([BP20, Discussion after Equation (11.2.6)])**.** *Let $\theta \in QC(G(F))$. Then for all $\mathrm{Re}(s) > 0$, the integral*

$$\int_{\Gamma(G,H)} D^G(x)^{1/2} c_\theta(x) \Delta(x)^{s-1/2} \mathrm{d}x$$

is absolutely convergent, and defines a continuous linear functional $m_{geom,s} \colon QC(G(F)) \to \mathbb{C}$.

Proof. If $L \subseteq G(F)$ is compact modulo conjugation, then there exists some seminorm ν_L on $QC(G(F))$ such that $D^G(x)^{1/2}\theta(x) \leq \nu_L(\theta)$ for $x \in L$,[2] and so we get a similar bound for $D^G(x)^{1/2} c_\theta(x)$ using Proposition 1.3.

Now $\Gamma(G,H)$ is itself compact modulo conjugation, so it remains to see that $\int_{\Gamma(G,H)} \Delta(x)^{s-1/2}\mathrm{d}x$ is absolutely convergent. We can compute that $\Delta(x) = |Nm_{E/F}(\det(1-x)|_{W''_x})|$ for any $x \in H_{ss}(F)$, and so

$$\int_{\Gamma(G,H)} \Delta(x)^{s-1/2}\mathrm{d}x = \sum_{T\in\mathcal{T}} \frac{1}{\#W(T)} \int_{T(F)} |\det(1-x)|_{W''_T}|^{2s-1}\mathrm{d}x.$$

It reduces to understanding each integral. In turn, this means seeing that $|\det(1-x)|_{W''_T}|^{2s-1}$ is locally integrable. The worst case is around $x = 1$, in which case $|\det(1-e^X)|_{W''_T}|$ is roughly $|\det X|_{W''_T}|$, and this is $\prod_i |\lambda_i(X)|$ for λ_i linearly independent functionals on $\mathfrak{t}(\bar{F})$[3]. Now we can see $\prod_i |\lambda_i(X)|^{2s-1}$ are locally integrable: it reduces to the 1-dim case since λ_i are linearly independent, and the 1-dim case is the fact that $\int_0^1 x^{2s-1}\mathrm{d}x$ converges. □

[2] This is part of the definition of quasi-characters.
[3] Since abelian lie algebras are upper triangular over $\bar{F}$, the eigenvalues are linear functionals

5. Proof of convergence as $s \to 0$

We recall that we will prove Theorem 2.2:

Theorem 5.1. *For each $\theta \in QC(G(F))$ the limit $m_{geom}(\theta) := \lim_{s\to 0^+} m_{geom,s}(\theta)$ exists, and m_{geom} defines a continuous functional on $QC(G(F))$.*

This proof is very long and technical. Roughly: one can control the behavior away from 1 by considering (G_x, H_x) and doing induction; one can control the behavior at 1 by rephrasing in terms of the Lie algebra and doing some very clever tricks we will see. The main trick will be a certain homogeneity property of the Lie algebra version m^{Lie}_{geom} of m_{geom}.

Step 0: LF spaces (such as $QC(G(F))$ satisfy a uniform boundedness principle (Banach–Steinhaus theorem): if a family of continuous operators converge pointwise, then it converges to a continuous operator, and the convergence is uniform on compact subsets. In particular, it suffices to prove $\lim_{s\to 0^+} m_{geom,s}(\theta)$ exists.

Step 1: By a partition of unity argument, we may assume that the support of θ is as small as we want.

Step 2: We can deal with cases $\theta \in QC(G(F))$ where $1 \notin \mathrm{Supp}(\theta)$ by considering (G_x, H_x, ξ_x) for $x \neq 1$. This is often called "semi-simple descent", and is done in [BP20, Equations 11.2.10-11.2.12].

Namely, consider θ with support close to a $x \in H_{ss}(F)$ with $x \neq 1$. Define $\Gamma(G_x, H_x) = \Gamma(G'_x, H'_x) \times \Gamma_{ani}(H''_x)$. (here Γ_{ani} is regular semisimple with compact centralizer), and the measure on $\Gamma_{ani}(H''_x)$ is given similarly as above, and then define

$$m_{x,geom,s} = \left(\theta_x \in QC(G_x(F)) \mapsto \int_{\Gamma(G_x,H_x)} D^{G_x}(y)^{1/2} c_{\theta_x}(y) \Delta^{G_x,H_x}(y)^{1/2-s} \mathrm{d}y\right).$$

We can compute that if $\theta_x = \theta'_x \otimes \theta''_x$ (note $QC(G_x(F)) = QC(G'_x(F)) \hat{\otimes} QC(G''_x(F))$), then

$$m_{x,geom,s}(\theta_x) = m^{(G'_x,H'_x)}_{geom,s}(\theta'_x) \times \int_{\Gamma_{ani}(H''_x)} D^{G''_x}(y)^{1/2} c_{\theta''_x}(y) \mathrm{d}y.$$

Now use induction hypothesis for the first term to prove $m_{x,geom}$ exists. Now if $\Omega_x \subseteq G_x(F)$ is a G-good small enough neighborhood of x and $\Omega = \Omega^G_x$, then if $\theta \in QC_c(\Omega)$ we can compute

$$m_{geom,s}(\theta) = m_{x,geom,s}((\eta^H_{x,G})^{1/2-s} \theta|_{G_x(F)})$$

for some factor $\eta^H_{x,G}$. We can check that $(\eta^H_{x,G})^{1/2}\theta|_{G_x(F)} \in QC_c(\Omega_x)$, and thus $m_{geom}(\theta)$ exists by the uniform boundedness principle.

Step 3: Now we zoom-in around 1. We do this by considering $m^{Lie}_{geom,s}$ and relating it to $m_{geom,s}$. $\Gamma^{Lie}(G,H)$ is defined similarly, in particular $\Gamma^{Lie}(G,H) \simeq \bigsqcup_{T\in\mathcal{T}} \mathfrak{t}_{\natural}(F)/W(T)$, and we consider

$$m^{Lie}_{geom,s} = \left(\theta \in QC_c(\mathfrak{g}(F)) \mapsto \int_{\Gamma^{Lie}(G,H)} D^G(X)^{1/2} c_\theta(X)\Delta(X)^{s-1/2}\mathrm{d}X\right).$$

For $\omega \subseteq \mathfrak{g}(F)$ a G-excellent open neighborhood of 0 and $\Omega = \exp(\omega)$, if $\theta \in QC_c(\Omega)$, then

$$m_{geom,s}(\theta) = m^{Lie}_{geom,s}((j^H_G)^{1/2-s}\theta_\omega)$$

and so again we only need to prove m^{Lie}_{geom} exists since $(j^H_G)^{1/2}\theta_\omega \in QC_c(\omega)$. This is explained in [BP20, Equations 11.2.14-11.2.15].

Step 4: If $\theta \in QC_c(\omega)$ and $0 \notin \mathrm{Supp}(\theta)$, then we can use the relation with $m_{geom,s}$ in reverse together with (2) to get that $m^{Lie}_{geom,s}(\theta)$ exists. This is done in [BP20, Equations 11.2.16-11.2.17].

Step 5: Now we improve this to all $\theta \in QC_c(\mathfrak{g}(F))$ and $0 \notin \mathrm{Supp}(\theta)$. This is [BP20, Equation 11.2.18].

The idea is that $\theta_\lambda(X) = \theta(\lambda^{-1}X)$ is in $\theta_\lambda \in QC_c(\omega)$ if $\lambda \in F^\times$ is sufficiently small. To relate them, we need the crucial

Lemma 5.2 (This is done in [BP20, Equations 11.2.19-11.2.20].)**.** *Denote $M_\lambda \in \mathrm{End}(QC_c(\mathfrak{g}(F)))$ given by $M_\lambda(\theta) = |\lambda|^{-\delta(G)/2}\theta_\lambda$. There exists $d > 0$ such that for all $\theta \in QC_c(\mathfrak{g}(F))$ and all $\lambda \in F^\times$ we have*

$$\lim_{s\to 0^+} m^{Lie}_{geom,s}((M_\lambda - 1)^d\theta) = 0.$$

Remark 5.3. As we will point out later, it will follow from the proof of the theorem that in fact $m^{Lie}_{geom}(M_\lambda\theta) = m^{Lie}_{geom}(\theta)$.

Proof. We start by computing that $D^G(X)^{1/2}c_{M_\lambda\theta}(X) = D^G(\lambda^{-1}X)c_\theta(\lambda^{-1}X)$. This follows from the limit formula Proposition 1.3.

So for example

$$m^{Lie}_{geom,s}(M_\lambda\theta) = \int_{\Gamma^{Lie}(G,H)} D^G(X)^{1/2}c_\theta(X)\Delta(\lambda X)^{s-1/2}\mathrm{d}(\lambda X) = \int_{\Gamma^{Lie}(G,H)} |\lambda|^{2sd_X} D^G(X)^{1/2}c_\theta(X)\Delta(X)^{s-1/2}\mathrm{d}X$$

where $d_X = \dim(T_X)$[4] (as $\Delta(X) = |Nm_{E/F}(\det(X|_{W_X''})|$ scales with λ). Note that m_{geom}^{Lie} is supposed to be a regularized integral

$$m_{geom}^{Lie}(\theta) = \int_{\Gamma^{Lie}(G,H)}^{*} D^G(X)^{1/2} c_\theta(X) \Delta(X)^{-1/2} \mathrm{d}X,$$

and the homogeneity property in the remark would follow if we could formally manipulate this integral by the above computation.

So

$$m_{geom,s}^{Lie}((M_\lambda - 1)^d \theta) = \int_{\Gamma^{Lie}(G,H)} (|\lambda|^{2sd_X} - 1)^d D^G(X)^{1/2} c_\theta(X) \Delta(X)^{s-1/2} \mathrm{d}X.$$

Now $||\lambda|^{2sd_X} - 1| \leq C \cdot s$, and so similarly to the proof of Proposition 4.1, it suffices to choose d so that

$$\lim_{s \to 0^+} s^d \int_{K \subseteq \mathfrak{t}(F)} \Delta(X)^{s-1/2} \mathrm{d}X = 0$$

for compact subsets K. Similar to before, this is choosing d large enough so that

$$\lim_{s \to 0^+} s^d \int_K \prod_{i=1}^{n} |\lambda(X)|^{2s-1} \mathrm{d}X = 0$$

when λ_i are linearly independent: if suffices $d > n$, since it reduces to the 1-dim case (note $\lim_{s \to 0^+} s^{1+\varepsilon} \int_0^1 x^{2s-1} = 0$). □

Step 6: Now we improve it one more time to all $\theta \in QC_{c,0}(\mathfrak{g}(F))$. Here the subscript 0 mean that $c_{\theta,\mathcal{O}}(0) = 0$ for all $\mathcal{O} \in \mathrm{Nil}_{reg}(\mathfrak{g})$. This is [BP20, Equation 11.2.21].

We claim that we can find θ_1, θ_2 with $0 \notin \mathrm{Supp}(\theta_2)$ and $\theta = (M_\lambda - 1)^d \theta_1 + \theta_2$. In fact, this is because $(M_\lambda - 1)$ is a bijection of $QC_0(\mathfrak{g}(F))$ with itself if $|\lambda| \neq 1$. Since $M_\lambda M_{\lambda^{-1}} = \mathrm{id}$, we can assume $|\lambda| > 1$, and then prove that $\sum_{n=0}^{\infty} (M_\lambda)^n \theta$ converges in $QC_0(\mathfrak{g}(F))$. In the p-adic case, this is because of the local expansion

$$M_\lambda^n \theta(X) = |\lambda|^{-n\delta(G)/2} \sum_{\mathcal{O} \in \mathrm{Nil}(\mathfrak{g})} c_{\theta,\mathcal{O}}(0) \hat{j}_{\mathcal{O}}(\lambda^{-n} X) = \sum_{\mathcal{O} \in \mathrm{Nil}(\mathfrak{g})} |\lambda|^{n \dim \mathcal{O}/2 - n\delta(G)/2} c_{\theta,\mathcal{O}}(0) \hat{j}_{\lambda^n \mathcal{O}}(X)$$

together with the fact that $\dim \mathcal{O} \leq \delta(G)$ with equality iff $\mathcal{O}$ is regular. In the real case this expansion is only assymptotic, so one needs to work a bit harder (see [BP20, Proposition 4.6.1]).

[4]Maybe there is a typo in Raphael's paper on page 270?

Remark 5.4. As Yiannis pointed out, in the p-adic case the above proof can be rephrased as follows: $QC_c(\mathfrak{g}(F))/QC_c(\mathfrak{g}(F))_{0\notin \mathrm{Supp}}$ is a finite dimensional vector space spanned by the germs of $\hat{j}_{\mathcal{O}}$ for $\mathcal{O} \in \mathrm{Nil}(\mathfrak{g})$, and their homogeneity property says that the action of M_λ is a sum of blocks, one for each $\dim \mathcal{O}$, and in each block it is a scalar times a permutation matrix. This scalar is 1 exactly when $\dim \mathcal{O} = \delta(G)$ (the regular case), and so the eigenvalues on $QC_{c,0}(\mathfrak{g}(F))/QC_c(\mathfrak{g}(F))_{0\notin \mathrm{Supp}}$ have absolute value $\neq 1$. So $M_\lambda - 1$ is invertible in $QC_c(\mathfrak{g}(F))/QC_{c,0}(\mathfrak{g}(F))$.

Step 7: Finally, it remains to prove that $m_{geom}(\theta)$ exists where $\theta = \varphi\hat{j}_{\mathcal{O}}$ for $\mathcal{O} \in \mathrm{Nil}_{reg}(\mathfrak{g})$ and φ a bump function. If G is not quasi-split these don't exist, and if G is, there are two of them: $\mathcal{O}_+$ and $\mathcal{O}_-$. So denote $\theta_\pm = \varphi\hat{j}_{\mathcal{O}_\pm}$. We use two tricks:

(a) $\sum_{\mathcal{O}\in \mathrm{Nil}_{reg}} \hat{j}_{\mathcal{O}} = i^G_{T_{qd}}(1)$ is parabolically induced from maximal torus of a Borel, and in particular it is supported on $\mathfrak{t}_{qd}(F)$. So in fact $\mathrm{Supp}(\theta_+ + \theta_-) \cap \Gamma^{Lie}(G,H) = \{0\}$, so

$$m_{geom,s}(\theta_+ + \theta_-) = c_{\theta_+}(0) + c_{\theta_-}(0) = 1$$

is constant with respect to s. This is explained in [BP20, Equations 11.2.23].

(b) If $\lambda \in F^\times \setminus Nm(E^\times)$, then $\lambda\mathcal{O}_\pm = \mathcal{O}_\mp$, so

$$(M_\lambda - 1)^d\theta_+ = (*)(\theta_+ - \theta_-) + (\text{supported away from } 0)\,.$$

This is explained after [BP20, Equations 11.2.23].

These imply that both $\lim_{s\to 0^+} m_{geom,s}(\theta_+ - \theta_-)$ and $\lim_{s\to 0^+} m_{geom,s}(\theta_+ + \theta_-)$ exist, so $\lim_{s\to 0^+} m_{geom,s}(\theta_+)$ and $\lim_{s\to 0^+} m_{geom,s}(\theta_-)$ also exist.

Remark 5.5. In fact, the above proof works "torus-by-torus", so $m_{geom} = \sum_{T\in\mathcal{T}} m^T_{geom}$ where $m^T_{geom} := \lim_{s\to 0^+} m^T_{geom,s}$. We computed that $m^{Lie,T}_{geom,s}(M_\lambda\theta) = |\lambda|^{2s\dim(T)} m^{Lie,T}_{geom,s}(\theta)$, and so taking limits we conclude that $m^{Lie}_{geom}(M_\lambda\theta) = m^{Lie}_{geom}(\theta)$.

6. Equivalence of geometric expansion and multiplicity formula

Recall the following two theorems Theorems 2.4 and 2.5:

Theorem 6.1. *$J(f) = J_{geom}(f) := m_{geom}(\theta_f)$ for all strongly cuspidal $f \in \mathcal{C}_{scusp}(G(F))$.*

Theorem 6.2. *$m(\pi) = m_{geom}(\pi) := m_{geom}(\theta_\pi)$ for all tempered π.*

Eventually, we will prove Theorem 6.1, and we want to deduce Theorem 6.2 as a corollary. In fact

Proposition 6.3 ([BP20, Proposition 11.5.1])**.** *Given a GGP triple (G, H, ξ), assume that we know that spectral expansion $J(f) = J_{spec}(f)$ and assume the induction hypothesis*

(HYP) *Theorems 6.1 and 6.2 hold for all (G', H', ξ') with $\dim(G') < \dim(G)$.*

Then Theorems 6.1 and 6.2 for (G, H, ξ) are equivalent.

For this we need a lemma

Lemma 6.4 ([BP20, Lemma 11.3.1])**.** *Let $L = L^{\mathrm{GL}} \times \tilde{G}$ be a Levi of G. We define m^L_{geom} a functional of $QC(L(F)) = QC(L^{\mathrm{GL}}(F))\hat{\otimes}QC(\tilde{G}(F))$ via extending*

$$m^L_{geom}(\theta^{\mathrm{GL}} \otimes \tilde{\theta}) := m_{geom}(\tilde{\theta}) \cdot c_{\theta^{\mathrm{GL}}}(1).$$

Then $m_{geom}(i^G_L(\theta)) = m^L_{geom}(\theta)$ for any $\theta \in QC(L(F))$.

Proof Sketch. We have the formula for the induction of a character

$$D^G(x)^{1/2} i^G_L(\theta)(x) = \sum_{x'} D^L(x')^{1/2}\theta(x')$$

where x' sums over representatives of the $L(F)$-conjugacy classes that meet the $G(F)$-conjugacy class of x. By the limit formula Proposition 1.3, this allow us to express $c_{i^G_L(\theta)}$ in terms of θ, and thus eventually $m_{geom,s}(i^G_L(\theta))$ in terms of $m^L_{geom}(\theta)$. □

Proof of Proposition 6.3. Recall that

$$J_{spec}(f) = \int_{\mathcal{X}(G)} D(\pi)\hat{\theta}_f(\pi)m(\bar{\pi})\mathrm{d}\pi = \sum_{\pi\in\mathcal{X}(G)_{ell}} D(\pi)\hat{\theta}_f(\pi)m(\bar{\pi}) + \int_{\mathcal{X}(G)_{ind}} D(\pi)\hat{\theta}_f(\pi)m(\bar{\pi})\mathrm{d}\pi.$$

By the induction hypothesis and the above lemma[5] we have $m(\bar{\pi}) = m_{geom}(\bar{\pi})$ for $\pi \in \mathcal{X}(G)_{ind}$, and so we can write

$$J_{spec}(f) = \int_{\mathcal{X}(G)} D(\pi)\hat{\theta}_f(\pi)m_{geom}(\bar{\pi})\mathrm{d}\pi + \sum_{\pi\in\mathcal{X}(G)_{ell}} D(\pi)\hat{\theta}_f(\pi)\left(m(\bar{\pi}) - m_{geom}(\bar{\pi})\right).$$

[5]We are also using that for θ^{GL}, having $c_{\theta^{\mathrm{GL}}}(1) = 1$ is equivalent to having a Whitakker model, which is true for tempered representations.

Note that

$$\int_{\mathcal{X}(G)} D(\pi)\hat{\theta}_f(\pi)m_{geom}(\bar{\pi})\mathrm{d}\pi = \int_{\mathcal{X}(G)} D(\pi)\hat{\theta}_f(\pi)m_{geom}(\theta_{\bar{\pi}})\mathrm{d}\pi = m_{geom}\left(\int_{\mathcal{X}(G)} D(\pi)\hat{\theta}_f(\pi)\theta_{\bar{\pi}}\mathrm{d}\pi\right) = m_{geom}(\theta_f).$$

Thus

$$J_{spec}(f) = J_{geom}(f) + \sum_{\pi\in\mathcal{X}(G)_{ell}} D(\pi)\hat{\theta}_f(\pi)\left(m(\bar{\pi}) - m_{geom}(\bar{\pi})\right).$$

Now Theorem 6.2 clearly implies Theorem 6.1. For the converse, fix a $\pi \in \mathcal{X}_{ell}(G)$. We would like to choose a $f \in \mathcal{C}_{scusp}(G(F))$ so that if $\pi' \in \mathcal{X}_{ell}(G)$, then $\hat{\theta}_f(\pi') \neq 0$ if and only if $\pi' = \pi$.

By orthogonality relations (due to Arthur), we have that if $\pi_1, \pi_2 \in \mathcal{X}_{ell}(G)$ then

$$\int_{\Gamma_{ell}(G)} D^G(x)\theta_{\pi_1}(x)\bar{\theta}_{\pi_2}(x)\mathrm{d}x = \begin{cases} D(\pi)^{-1} & \text{if } \pi_1 = \pi = \pi_2, \\ 0 & \text{otherwise.} \end{cases}$$

Moreover all $\pi \in \mathcal{X}_{ind}(G)$ have characters that vanish on $G_{reg,ell}(F)$, so the plan is to take f such that $\theta_f(x) = \theta_\pi(x)$ for all $x \in \Gamma_{ell}(G)$. In the p-adic case, we can take such an $f \in \mathcal{S}_{scusp}(G(F))$, but in the real case we need to be more careful, since we can only guarantee such a f as a limit of strongly cuspidal functions (see [BP20, Corollary 5.7.2(i)], and [BP20, Corollary 5.7.2(iv)] for the end of the proof in the real case). □

References

[BP20] Raphaël Beuzart-Plessis. A local trace formula for the Gan-Gross-Prasad conjecture for unitary groups: the Archimedean case. *Astérisque*, (418):ix+305, 2020.